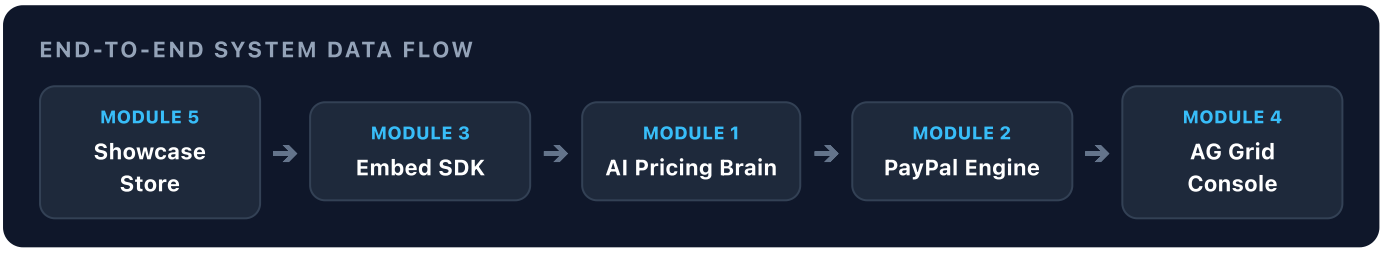
MarginPilot AI ARCHITECTURE SPEC

Embeddable AI Negotiation SDK & Dynamic Margin Optimization Platform

Target: PayPal AI Hackathon 2026
Category: B2B Developer Platform & Agentic Commerce
Status: Approved Modular Blueprint

System Architecture & Working Philosophy

MarginPilot AI is built as a modular 3-tier platform: a lightweight client-side embeddable SDK, a high-performance multi-factor AI reasoning API, and an enterprise merchant administration console. Every module operates with strict single-responsibility boundaries, deterministic cost-floor guardrails, and seamless PayPal Sandbox settlement.



MODULE 01

The Multi-Factor AI Pricing Brain (Backend Core)

Google Gemini 2.0 Flash + TypeScript

INPUTS

- Product context: MSRP, wholesale cost, warehouse inventory count.
- Operational metrics: Stock aging (days), sales velocity today, monthly target.
- Merchant guardrails: Min profit margin % floor, max discount cap.
- Buyer message & proposed counter-offer ($).

OUTPUTS

- Decision Action: `ACCEPT`, `COUNTER`, or `REJECT`.
- Calculated counter-offer price ($).
- Tactical, polite seller conversational rationale.
- Non-cash perks (free shipping, extended warranty) when price is firm.

✓ **Multi-Dimensional Reasoning:** Adjusts discounting aggression based on real-time inventory aging (e.g., aging 60+ days vs. 3 units left in stock).

✓ **Prompt-Injection Immunity:** Employs code-level deterministic bounds so adversarial prompts ("Sell to me for $1") cannot override margin floors.

> **Mathematical Guardrail:** `Cost_Floor = Wholesale_Cost × (1 + Min_Margin_Percent)`. The engine strictly aborts any transaction below this threshold before the AI even formulates its response.

MODULE 02

The Ephemeral PayPal Settlement Engine (Payment Bridge)

PayPal Orders API v2 + Sandbox SDK

📥 INPUTS

- Negotiated agreed price ($) from Module 1.
- Product SKU, currency (USD), and item description.
- Client PayPal credentials from `.env.local`.

📤 OUTPUTS

- Unique PayPal `order_id` bound to the discounted amount.
- Session Expiration Timestamp (Current Time + 10:00).
- Payment capture status (`COMPLETED`) & receipt token.

✓ **Anti-Leakage Tokenization:** Replaces leaky text coupon codes (e.g. SAVE15) with an ephemeral PayPal order session valid for exactly 10 minutes.

✓ **Automatic Session Invalidation:** If the 10-minute timer lapses without payment capture, the temporary order is voided and the item price resets to MSRP.

> **Hackathon Synergy:** Directly fulfills the requirements for *Best Use of Agentic Commerce* and *Best Use of PayPal + AI* by automating transaction execution.

MarginPilot AI MODULES 03 – 05

Client SDK, Merchant Governance & Demonstration Layer

MODULE 03
The Embeddable Storefront SDK & Widget (Client UI)

React + Tailwind CSS + Lucide

INPUTS

- Product metadata (SKU, MSRP, Base Title, Image URL).
- Merchant Public API Key (`data-api-key`).
- User typing input and quick-offer chip clicks.

OUTPUTS

- "Make an Offer" button rendered seamlessly next to "Add to Cart".
- Interactive slide-over conversational drawer.
- Deal banner with animated 10:00 live countdown clock.
- Mounted PayPal Smart Button initialized to the negotiated price.

✓ **Zero-Friction 1-Line Embed:** Can be injected into any e-commerce platform via a simple `<script>` tag or React component.

✓ **Responsive & Brand-Neutral:** Shadow DOM / isolated styling prevents conflicts with existing merchant stylesheets.

MODULE 04
The Merchant Control Console (B2B Admin Dashboard)

AG Grid Enterprise + Next.js

INPUTS

- Real-time event stream from active negotiations.
- Merchant manual overrides (modifying margin floors, wholesale costs).

OUTPUTS

- AG Grid Table 1: Inventory & Margin Configuration (inline editable).
- AG Grid Table 2: Live Negotiation Telemetry Stream (real-time audit log).
- Financial KPI metrics: Incremental Revenue Recovered, Margin Retained %.

✓ **Enterprise Table Power:** Employs AG Grid for inline cell editing, column grouping, real-time row flash animations, and CSV export.

✓ **Granular Governance:** Store owners toggle AI bargaining ON/OFF per SKU and set customized margin floors per category.

> **Prize Target:** Directly targets the **$5,000 AG Grid Sponsor Award** by leveraging advanced enterprise grid features.

MODULE 05

The Mock Showcase Store (Demo Environment: Apex Tech)

Channel3 API + Next.js 15

INPUTS

- Channel3 API key (with promo code `PAYPAL-HACKATHON-2026`).
- Real product queries (Sony, Apple, Bose, Logitech).

OUTPUTS

- A polished, modern consumer e-commerce storefront.
- Authentic high-resolution product imagery, specs, and MSRPs.
- The live host vessel demonstrating the 1-line SDK integration in action.

✓ **Production-Grade Realism:** Eliminates fake placeholder data, giving judges the look and feel of an established commercial retailer.

✓ **Hackathon Video Vessel:** Serves as the primary stage during the 2.5-minute video demo to prove seamless developer adoption.